VialSense Runbook

This runbook explains how to install, test, and run the VialSense prototype. Commands start from the project root. Python 3.10 or newer is recommended.

1. Create the Python environment

macOS / Linux:

```
python3 -m venv .venv
source .venv/bin/activate
python -m pip install --upgrade pip
python -m pip install -r requirements.txt
python -m pip install -r requirements-dashboard.txt
```

Windows PowerShell:

```
py -3 -m venv .venv
.\.venv\Scripts\Activate.ps1
python -m pip install --upgrade pip
python -m pip install -r requirements.txt
python -m pip install -r requirements-dashboard.txt
```

2. Run the tests

macOS / Linux:

```
PYTHONPATH=src python -m pytest -q
```

PowerShell:

```
$env:PYTHONPATH = "src"
python -m pytest -q
```

3. Rebuild the saved Dataset 2 results

Send output to a review directory so the checked-in results are not overwritten:

```
PYTHONPATH=src python -m vialsense.corrected_cli build --output-dir /tmp/vialsense-review
```

This rebuilds the RGB/HSV timelines and provisional anomaly profiles from the saved Dataset 2 measurements and fill labels.

4. Replay automatic live tracking on Dataset 2 Video 5

Video 5 is not included in the repository. With a local copy of the original MP4:

```
PYTHONPATH=src python -m vialsense.live_cli run \
  --video /path/to/vial_fill_video_5.mp4 \
  --recipes "5x3,6x3,7x3,8x3,9x3,10x3,11x3,12x3,13x3,14x3" \
  --fps 5 --no-window --output-dir /tmp/vialsense-live-review
```

This detects vials, assigns recipes from the operator queue, estimates fill progress from the detected liquid rise, calculates RGB/HSV, and scores usable readings with the anomaly detector. The output folder includes frame readings, per-vial summaries, events, and a run summary. Use the correct queue for the video; queue misalignment shifts recipe assignments.

PowerShell uses `$env:PYTHONPATH = "src"` on its own line before the command.

5. Launch the operator dashboard

```
python -m streamlit run app.py
```

Allow camera access in the browser. Select **Automatic vial tracking**, enter the recipe queue in production order, and include the preceding recipe if it is known. Use **Skip missed vial** or **Undo queue step** if vial counting loses alignment. **Manual fallback** lets an operator provide recipe and fill progress when automatic tracking is not reliable.

If the recipe order is unknown, clear the queue field. The detector and vial tracker continue to display RGB/HSV readings, but the dashboard will not assign recipes, estimate recipe-based fill percentages, or produce anomaly scores. If the vial arrives with liquid already visible, it reports **Fill start not observed**; that vial is not scored, including in its final summary. This keeps the system from treating an assumed early-fill stage as measured progress.

The automatic camera view also shows recipe-change warnings, the latest completed vial's verdict and explanation, and a table of completed vials. Use the table to review `PASS`, `FLAG`, `NO DATA`, and `NO RECIPE` outcomes during the run; each `FLAG` is a request for review, not a confirmed contamination.

The liquid finder expects the Dataset 2 camera geometry. For a different camera view, create a calibration JSON with `vialsense.live_cli calibrate`, inspect the generated preview, and upload the JSON in the dashboard. Calibrate from a machine where the target camera is accessible. If the holder is not aligned in the corrected preview, do not rely on the readings.

6. Run the saved integration workflow

Saved detector readings and Video 5 labels can be replayed without the original video:

```
PYTHONPATH=src python -m vialsense.integration_cli saved \
  --output-dir /tmp/vialsense-saved-review
```

For full setup details and limitations, see DASHBOARD_SETUP.md, LIVE_MODE.md, and the project README.

What the results mean

Recipe profiles and anomaly scores are exploratory. Dataset 2 does not provide confirmed contamination labels, so `FLAG` or `review_unusual` requests an operator review and does not confirm contamination or product failure. RGB/HSV values depend on lighting, camera alignment, reflections, fill level, and the finder's selected region. The current tracker was replayed on Video 5 and on an unlabeled CSI recording; the CSI replay supports detector/tracker checks only and does not provide recipe-based anomaly results. The updated workflow has not yet been retested with a live camera on the physical CSI station.